From Relation to Structure: Spatial-Semantic Guidance and Structure Refinement Network for Multimodal Remote Sensing Segmentation

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***Abstract*—Semantic segmentation is a fundamental task in remote sensing image (RSI) interpretation, aiming at pixel-wise classification of land cover. Recently, multimodal RSI semantic segmentation has attracted significant attention for its ability to alleviate the information bottleneck inherent in unimodal methods. However, existing multimodal methods often emphasize feature alignment and complementarity while overlooking the inherent coexistence and mutually exclusive relationships among land covers, frequently resulting in fragmented predictions. Moreover, achieving fine-grained boundary segmentation remains a crucial challenge. To address these challenges, we propose R2SNet, a spatial-semantic guidance and structure refinement network for multimodal RSI segmentation. The core innovation of R2SNet lies in a progressive optimization strategy that guides from semantic relations modeling to spatial structure refinement via three collaborative modules. Specifically, the asymmetric dual-branch (ADB) encoder performs modality-specific extraction of semantic and geometric features to mitigate computational redundancy and modal interference. Subsequently, the spatial-semantic guided cross-modal fusion (SGCF) module explicitly embeds spatial-semantic priors into the fusion process to suppress unreasonable relationships. Finally, the progressive structure refinement (PSR) decoder aggregates multiscale contexts and employs an eight-directional selective scanning mechanism to enhance structural integrity and refine boundaries. Extensive experiments on the ISPRS Vaihingen, ISPRS Potsdam, DDHR, and WHU-OPT synthetic aperture radar (SAR) datasets demonstrate that R2SNet outperforms existing state-of-the-art methods, particularly regarding semantic consistency and boundary precision. The code will be available at: https://github.com/bnu-wgy/R2SNet**

***Index Terms*—Multimodal remote sensing, semantic segmentation, spatial-semantic relationship.**

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I. Introduction

SEMANTIC segmentation plays a crucial role in remote sensing image (RSI) interpretation, aiming to assign specific category labels to each pixel. As a fundamental task for advanced Earth observation analysis, it provides essential support for urban planning [1], land cover mapping [2], postdisaster damage assessment [3], and many other fields [4], [5].

Distinct from natural images, RSI spans vast geographic areas and encompasses complex backgrounds and land cover distributions [6]. For a long time, semantic segmentation methods that utilize optical RSI have been dominant [7]. However, constrained by the passive imaging mechanism, optical sensors primarily measure reflected radiance and capture surface spectral characteristics [8]. Consequently, optical RSI is not only susceptible to variations in illumination and weather conditions but also lacks elevation information. With the rapid advancement of satellite and airborne sensor technologies, multimodal RSI, encompassing optical, light detection and ranging (LiDAR), and synthetic aperture radar (SAR), has experienced explosive growth. These multimodal RSIs provide complementary information for Earth observation [9]. For example, optical imagery offers rich texture and spectral detail, digital surface model (DSM) provides elevation and geometric information of land covers [10], while SAR imagery enables all-weather imaging and the ability to penetrate clouds and fog [11]. Currently, leveraging the complementarity of multimodal RSI to overcome information bottlenecks has proven to be an effective strategy to improve segmentation accuracy in complex remote sensing scenes.

Extensive research has focused on multimodal fusion strategies to enhance the effective utilization of multimodal RSI [12], [13]. Early research predominantly employed additive fusion, such as PACSCNet [14], OPDNet [15], and others [16]. While these methods are direct and computationally efficient, they implicitly assume that different modalities contribute equally to downstream tasks [17]. This assumption overlooks variations in noise distribution and information redundancy between different modalities. Therefore, additive fusion often propagates noise or interference from certain modalities into the fused features, contaminating the effective information of complementary modalities and hindering the exploitation of multimodal complementarity and synergy [18]. To mitigate

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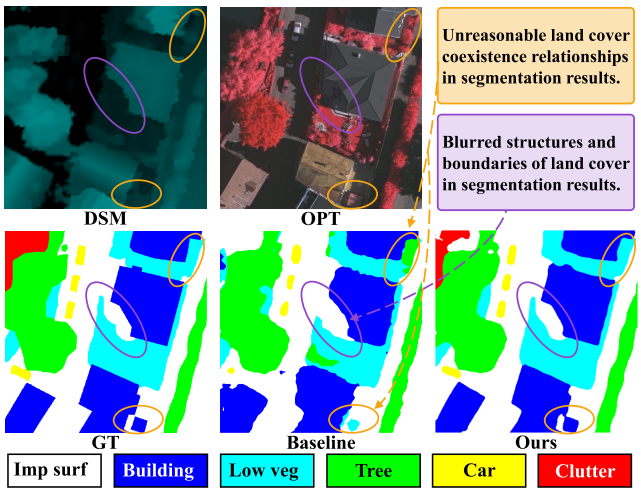


Fig. 1. Our observations and basic motivation. As highlighted by the orange ellipse, the baseline method ignores inherent semantic relationships (co-occurrence and mutual exclusion), leading to illogical predictions and fragmented semantics. Meanwhile, the purple ellipse indicates unclear structure with blurred boundaries. R2SNet yields structurally complete segmentation results with fine-grained boundaries.

these limitations, researchers have developed fusion strategies based on attention and selection mechanisms, such as FTransUNet [19], StitchFusion [20], and others [21]. These methods prioritize modeling nonlinear cross-modal correlations over direct concatenation. By introducing feature selection and weighting mechanisms at the channel or spatial dimensions, these methods adaptively recalibrate the contribution weights of different modalities based on contextual information. This emphasizes critical details while suppressing irrelevant or low-quality information [22], [23]. This cross-modal interaction significantly enhances the robustness and discriminative performance of fused features, enabling the model to deeply extract multimodal RSI and achieve outstanding performance in complex remote sensing scenes.

However, in this line of research, existing works have effectively advanced multimodal fusion while still facing two major challenges

1) The inherent coexistence and mutually exclusive relationships between land covers in RSI are overlooked. As shown in Fig. 1, trees rarely adjoin buildings, and small rectangular regions are more likely to be buildings than low vegetation. Neglecting this high-level semantic guidance and these constraints makes models prone to generating counterintuitive predictions, leading to semantic fragmentation and ambiguity in results.
2) Fine-grained boundary segmentation remains challenging in complex remote sensing scenarios.

Motivated by the discussions above, this article proposes a spatial-semantic guidance and structure refinement network (R2SNet) to establish a progressive optimization strategy from semantic relationship guidance to spatial structure refinement. Specifically, R2SNet employs three collaborative modules to address cross-modal interference, semantic deficiency, and boundary blurring. First, the asymmetric dual-branch (ADB) encoder is designed to extract modality-specific multiscale semantic and geometric features. By fully leveraging the representational characteristics of each modality, it effectively mitigates information redundancy and modal interference at the source. Subsequently, targeting semantic relationships, the spatial-semantic-guided cross-modal fusion (SGCF) module is introduced. This module explicitly embeds spatial-semantic priors with coexistence and mutual exclusion constraints into the cross-modal fusion process to suppress unreasonable predictions. Finally, the progressive structure refinement (PSR) decoder is developed. This decoder aggregates multiscale features and employs an eight-directional selective scanning mechanism to capture directional dependencies, thereby improving structural integrity and refining boundaries.

To validate the effectiveness of R2SNet, we conducted extensive experiments on four benchmark datasets: ISPRS Vaihingen, ISPRS Potsdam, DDHR, and WHU-OPT-SAR. These datasets cover two representative multimodal combinations, optical-DSM and optical-SAR. The experiments indicate that R2SNet outperforms existing state-of-the-art unimodal and multimodal methods, particularly in semantic coherence and fine-grained boundary delineation. In summary, the contributions of this article are as follows.

1) We propose R2SNet for multimodal RSI semantic segmentation. R2SNet establishes a progressive optimization strategy from semantic relationship modeling to spatial structure refinement. The ADB encoder is introduced to extract modality-specific semantic and geometric representations, providing a reliable foundation for progressive fusion and structure refinement.
2) We design the SGCF module to embed spatial-semantic priors into multimodal feature fusion. By jointly modeling semantic distributions and spatial positions, SGCF constructs a guidance bias for cross-modal attention, suppressing unreasonable semantic associations and enhancing semantic coherence and logical consistency.
3) We develop the PSR decoder to improve structural integrity and boundary delineation. Progressive multiscale fusion is used to integrate deep semantics with shallow spatial details, while an eight-directional selective scanning mechanism captures axial and diagonal structural dependencies, improving the representation of complex land-cover structures and fine-grained boundaries.

The remainder of this article is organized as follows. Section II reviews related work on both unimodal and multimodal RSI semantic segmentation. Section III details the overall architecture of R2SNet and its constituent components. Section IV describes experimental preparations, comparative experiments, and ablation studies, followed by analysis and discussion of the results. Section V discusses the limitations of R2SNet. Finally, Section VI summarizes the research conclusions.

II. Related Work

To better position this work and clarify its innovations, a systematic review of relevant research is necessary. First, we reviewed the technical foundations and challenges in unimodal RSI semantic segmentation. We focused on analyzing its inherent limitations in information extraction, effective global context modeling, and fine-grained structure recovery,

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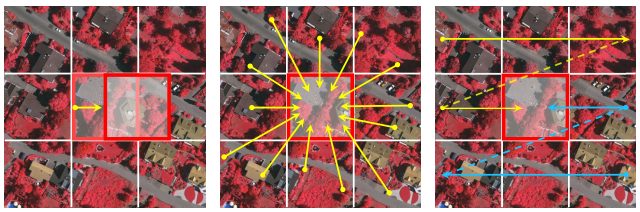


Fig. 2. Three modeling mechanisms of mainstream methods. These structures are conceptual examples with variations in specific implementations. (a) CNN-based. (b) Transformer-based. (c) Mamba-based.

providing preliminary support for the necessity of multimodal methods. Subsequently, we delved into the evolution of multimodal RSI semantic segmentation, particularly the contributions existing methods have made in multimodal fusion. Building upon this, we identified bottlenecks in current methods regarding the effective capture of semantic relationships between land covers, as well as the representation of fine-grained structures and boundaries in complex scenes.

A. Unimodal RSI Semantic Segmentation

With the rapid development of deep learning, significant progress has been made in the field of RSI semantic segmentation. Fig. 2 illustrates the modeling mechanisms of mainstream methods. Convolutional neural networks (CNNs) have demonstrated outstanding performance with their powerful nonlinear feature extraction capabilities, maintaining a long-standing dominance [24], [25]. However, restricted by receptive fields, CNN-based methods are often inadequate in capturing global semantic dependencies in large-scale remote sensing scenes [26]. Subsequently, methods such as DeepLabv3+ [27] and A2FPN [28] adopted strategies including dilated convolution [29] and attention mechanisms [30] to mitigate this issue, but they still struggle to resolve it fundamentally.

Addressing this limitation, the vision transformer (ViT) and its variants were developed [31], [32]. In contrast to CNN-based methods, Transformer explicitly models long-range correlations through self-attention mechanisms, capturing global contextual dependencies [33], [34]. Theoretically, Transformer possesses a global receptive field during feature extraction, effectively compensating for the local limitations inherent in convolutional operations. Subsequently, hybrid architectures that combine the local detail extraction capabilities of CNN with the global modeling advantages of Transformer have attracted significant interest [35]. Recently, Mamba-based methods [36] introduced selective scanning mechanisms, achieving efficient long-range dependency modeling while maintaining linear computational complexity [37], [38], [39]. This offers new perspectives for processing high-resolution RSI with large fields of view. Additionally, visual foundation models have garnered attention, particularly the Segment Anything Model (SAM) [40] and its variant MobileSAM [41], which have demonstrated outstanding performance. Pretrained on millions of natural images, SAM exhibits strong zero-shot generalization capabilities, generating high-quality segmentation masks from prompts. To further enhance computational efficiency, MobileSAM reduces parameters by over 60× while maintaining comparable performance [42]. Compared to other variants like FastSAM [43], which prioritize extreme speed at the expense of performance, it strikes a better balance between efficiency and segmentation quality. While SAM demonstrates outstanding performance in natural image processing, it requires dedicated adaptation for effective application to RSI segmentation due to substantial domain gaps [44], [45].

Despite breakthroughs in feature extraction achieved by the aforementioned methods, unimodal RSI segmentation remains constrained by insufficient information content. Optical RSI is susceptible to weather variations, shadow occlusion, and lacks elevation information for land covers [46]. When dealing with complex scenes characterized by unstable interclass and intraclass variance, relying solely on unimodal visual features typically proves insufficient for accurate semantic discrimination. Therefore, richer semantic representations and contextual information are indispensable to meet the demands of complex remote sensing scenes.

B. Multimodal RSI Semantic Segmentation

Driven by advancements in satellite and aerospace technology, remote sensing encompasses diverse modalities with distinct physical properties, providing critical complementary information for semantic segmentation [8], [47]. For example, the near-infrared (NIR) band effectively distinguishes crop types from natural vegetation by capturing unique spectral reflectance characteristics of vegetation [48]. SAR offers all-weather imaging capabilities, providing surface texture and structural information unaffected by cloud cover or lighting conditions [49]. DSM provides elevation data for terrain, enabling the differentiation of objects with similar spectral characteristics but varying heights [50]. To overcome the inherent limitations of the optical modality, multimodal RSI semantic segmentation integrating such heterogeneous data has become a dominant research trend.

Multimodal RSI contains abundant information, yet efficiently integrating these heterogeneous data presents a significant challenge. Early research primarily focused on linear aggregation. PACSCNet [14] employs a symmetric cascaded architecture that progressively coordinates features across adjacent layers to preserve spatial details in multimodal RSI segmentation. However, such direct aggregation often causes noise from a specific modality to be propagated into the fused features, undermining feature discriminability. To address this challenge, fusion strategies based on attention and gating mechanisms have emerged. SAGate [51] mitigates deep feature noise by introducing separate gating mechanisms within a cross-modal-guided encoder. AMIANet [21] employs attention mechanisms to dynamically balance and fuse multimodal RSI, enabling adaptive intermodal interaction. FTransUNet [19] combines CNN and Transformer to capture both local texture information and global semantic associations simultaneously. Meanwhile, CSFAFormer [52] dynamically adjusts interaction weights to leverage complementary advantages. These methods effectively align features across different spatial resolutions and mitigate domain gaps by modeling nonlinear dependencies between modalities. By utilizing dynamically generated weight masks for adaptive recalibration, they actively enhance complementary information

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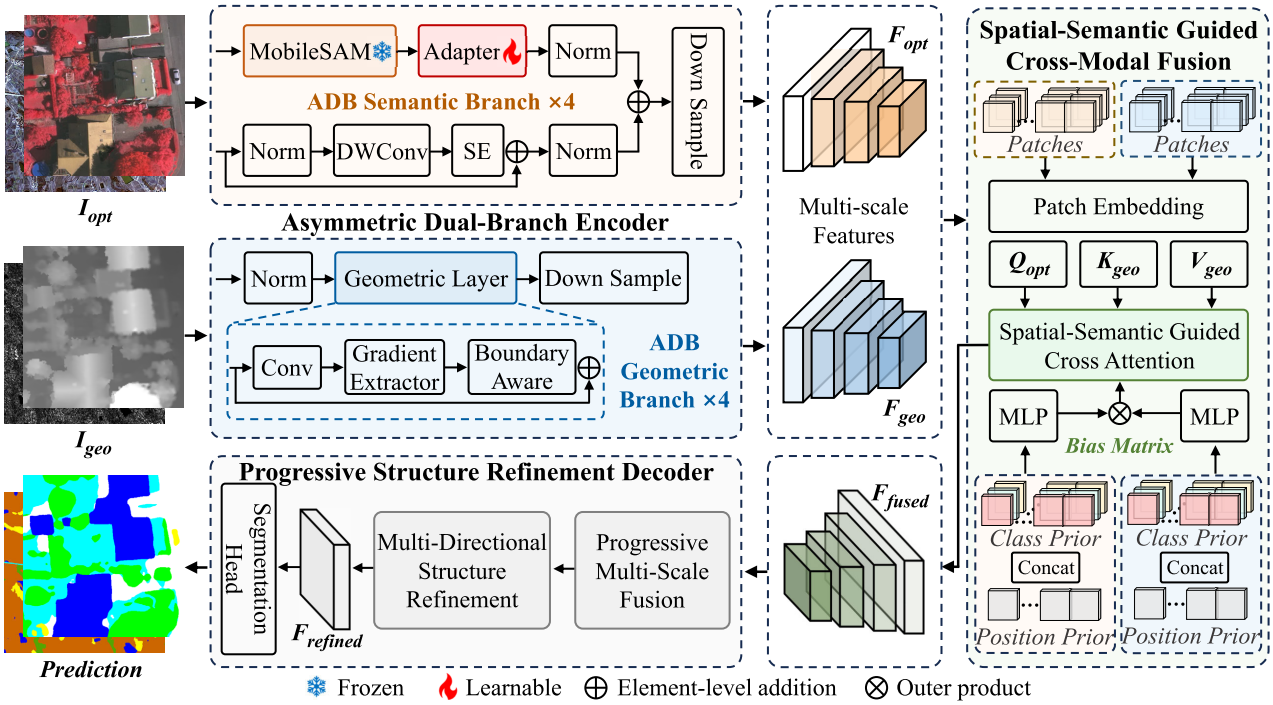


Fig. 3. Architecture of the proposed R2SNet. This network consists of the ADB encoder, SGCF module, and PSR decoder.

while suppressing redundancy, improving collaborative representation [53], [54].

Extensive efforts in multimodal fusion have achieved significant breakthroughs in multimodal RSI semantic segmentation. However, numerous fusion strategies still treat segmentation merely as a pixel-level probability fitting task, neglecting the inherent coexistence and mutually exclusive relationships among land covers. This often leads to semantically fragmented and ambiguous segmentation results. Therefore, it is necessary to develop models capable of effectively capturing the relationships between land covers.

III. Method

A. Architecture Overview

As illustrated in Fig. 3, the proposed R2SNet is composed of three core components: ADB encoder, SGCF module, and PSR decoder. Specifically, R2SNet takes paired optical image $\mathbf{I}_{\text{opt}}$ and DSM (or SAR) image $\mathbf{I}_{\text{geo}}$ as input. Given that the optical image offers rich semantic information while DSM (or SAR) excels at capturing geometric structure and boundaries, $\mathbf{I}_{\text{opt}}$ is treated as the primary modality and $\mathbf{I}_{\text{geo}}$ as the auxiliary modality. The ADB encoder adopts modality-specific branches to extract multiscale semantic features $\mathbf{F}_{\text{opt}}$ and geometric features $\mathbf{F}_{\text{geo}}$, minimizing cross-modal redundancy at the source. Subsequently, the extracted multimodal features are fed into the SGCF module. This module explicitly encodes spatial-semantic priors that encapsulate the coexistence and mutually exclusive relationships of land covers into a bias matrix. This bias matrix is embedded within the cross-attention to guide cross-modal fusion, explicitly encoding spatial-semantic relationships that characterize the land cover patterns of the scene. The fused multiscale features $\mathbf{F}_{\text{fused}}$ are then delivered to the PSR decoder for scale aggregation and refinement. PSR progressively integrates multiscale features and introduces an eight-directional selective scanning to further refine structural integrity and boundaries. Finally, the segmentation head maps the refined features $\mathbf{F}_{\text{refined}}$ into the label space to produce accurate segmentation results.

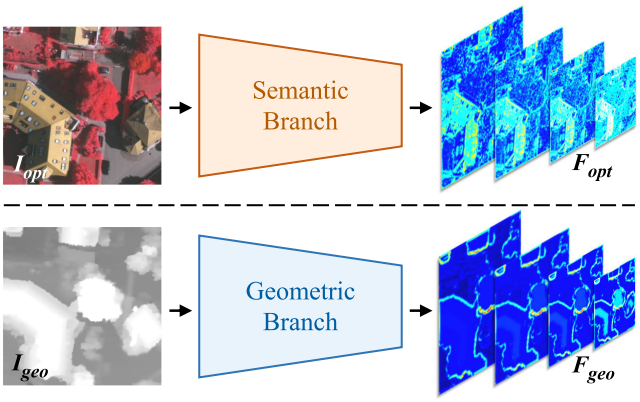


Fig. 4. Components of the ADB encoder include the semantic branch and the geometric branch. Each branch outputs feature maps with four scales.

B. ADB Encoder

For multimodal feature extraction, existing methods often employ homogeneous architectures to process distinct modalities. Although such siamese structures are simple and unified, their similar processing patterns tend to introduce redundancy and weaken the effective capture of features from different modalities. For modality-specific feature extraction, we introduce the ADB encoder, as shown in Fig. 4. This

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encoder comprises a semantic branch dedicated to the semantic interpretation of the optical modality and a geometric branch responsible for the extraction of the geometric structure from the auxiliary modality, respectively.

1) Semantic Branch: The semantic branch employs a pretrained MobileSAM with frozen parameters to extract semantic features from $\mathbf{I}_{\text{opt}}$. To bridge the domain gap between the pretraining domain and remote sensing scenes, we introduce a lightweight adapter consisting of a linear projection, feature normalization, and nonlinear activation. This adapter efficiently maps general visual features to suit the current task, achieving semantic adaptation

$$\mathbf{F}_{\text{global}} = \text{ReLU}\left(\text{BN}\left(\text{Conv}_{1\times1}\left(\text{MobileSAM}\left(\mathbf{I}_{\text{opt}}\right)\right)\right)\right). \quad (1)$$

To accommodate the extraction of local details, a lightweight pipeline incorporating depthwise convolution (DWConv) and a squeeze-excitation (SE) block [30] is parallelized in this branch

$$\mathbf{F}_{\text{local}} = \text{SE}\left(\text{DWConv}\left(\mathbf{I}_{\text{opt}}\right)\right). \quad (2)$$

Subsequently, the global feature $\mathbf{F}_{\text{global}}$ and local feature $\mathbf{F}_{\text{local}}$ are concatenated and downsampled to generate the multiscale semantic features $\mathbf{F}_{\text{opt}}$

$$\mathbf{F}_{\text{opt}} = \text{Down}_l\left(\text{Conv}_{1\times1}\left(\text{Concat}\left(\mathbf{F}_{\text{global}}, \mathbf{F}_{\text{local}}\right)\right)\right) \quad (3)$$

where $\text{Down}_l(.)$ denotes the downsampling operation at the lth scale, and $\mathbf{F}_{\text{opt}} = \{\mathbf{F}^l_{\text{opt}}\}^4_{l=1}$ represents the generated multiscale semantic features. The semantic branch effectively achieves complementary advantages between global context and local details, ensuring robust feature extraction under complex land cover distributions.

2) Geometric Branch: The geometric branch employs a gradient extractor and a boundary-aware block to capture spatial discontinuities and structural boundaries. First, to perceive the directional characteristics of terrain undulations, finite differences are employed to compute the horizontal and vertical geometric gradients $\nabla_x\mathbf{I}_{\text{geo}}$ and $\nabla_y\mathbf{I}_{\text{geo}}$

$$\begin{cases}\nabla_x\mathbf{I}_{\text{geo}}(i,j) = \mathbf{I}_{\text{geo}}(i,j+1) - \mathbf{I}_{\text{geo}}(i,j)\\ \nabla_y\mathbf{I}_{\text{geo}}(i,j) = \mathbf{I}_{\text{geo}}(i+1,j) - \mathbf{I}_{\text{geo}}(i,j).\end{cases} \quad (4)$$

These geometric gradients not only characterize the terrain slope in DSM data but also effectively capture abrupt changes in radiometric intensity within SAR images induced by corner reflections. To adaptively weight directional gradients and suppress noise, we introduce learnable directional parameters λ_x and λ_y. Subsequently, by concatenating the input, weighted gradient, and gradient magnitude $\mathbf{M}$, the composite feature vector $\mathbf{V}$ is constructed

$$\mathbf{V} = \text{Concat}\left(\mathbf{I}_{\text{geo}}, \lambda_x\nabla_x\mathbf{I}_{\text{geo}}, \lambda_y\nabla_y\mathbf{I}_{\text{geo}}, \mathbf{M}\right) \quad (5)$$

where $\mathbf{M}$ is calculated as the Euclidean norm of $\nabla_x\mathbf{I}_{\text{geo}}$ and $\nabla_y\mathbf{I}_{\text{geo}}$. Subsequently, a 1×1 convolution reduces its dimensionality to generate the gradient feature $\mathbf{F}_{\text{grad}}$.

Furthermore, the boundary-aware block enhances the capability to capture closed contours and subtle boundaries. This block employs an isotropic second-order differential operator to pinpoint regions of abrupt change by capturing second-order variations in image intensity. Specifically, we utilize a convolutional layer with learnable weights, where the weight matrix $\mathbf{W}$ is initialized with the Laplacian operator $\mathbf{W}_{\text{Lap}}$

$$\mathbf{W}_{\text{Lap}} = \begin{bmatrix}0 & 1 & 0\\ 1 & -4 & 1\\ 0 & 1 & 0\end{bmatrix}. \quad (6)$$

Finally, multiscale geometric features are generated by applying a Laplacian-initialized convolution to $\mathbf{F}_{\text{grad}}$ followed by scale-specific downsampling

$$\mathbf{F}_{\text{geo}} = \text{Down}_l\left(\text{Conv}_{\text{Lap}}\left(\mathbf{F}_{\text{grad}}\right)\right). \quad (7)$$

The geometric branch explicitly models spatial discontinuities using learnable directional gradients and Laplacian priors. The generated multiscale geometric features $\mathbf{F}_{\text{geo}}$ are spatially aligned with the corresponding semantic features $\mathbf{F}_{\text{opt}}$ at each scale, establishing a unified multiscale representation for subsequent cross-modal fusion.

Algorithm 1 Process of SGCF Module

Inputs: Optical features $\mathbf{F}_{\text{opt}}$, Geometric features $\mathbf{F}_{\text{geo}}$
Output: Fused multimodal features $\mathbf{F}_{\text{fused}}$
Flatten $\mathbf{F}_{\text{opt}}$, $\mathbf{F}_{\text{geo}}$ into token sequences $\mathbf{T}_{\text{opt}}$, $\mathbf{T}_{\text{geo}}$
for each modality $m \in \{\text{opt}, \text{geo}\}$ **do**
  Project tokens to semantic space to obtain $\mathbf{P}_m$ by (8)
  Construct normalized spatial coordinate grid $\mathbf{G}$ by (9)
  Fuse priors to generate embedding $\mathbf{E}_m$ by (10)
end for
Compute logical compatibility bias matrix $\mathbf{B}_{\text{guide}}$ by (11)
Generate $\mathbf{Q}_{\text{opt}}$ from $\mathbf{F}_{\text{opt}}$, $\mathbf{K}_{\text{geo}}$ and $\mathbf{V}_{\text{geo}}$ from $\mathbf{F}_{\text{geo}}$
Inject $\mathbf{B}_{\text{guide}}$ into attention map to obtain $\mathbf{F}_{\text{att}}$ by (12)
Calculate $\mathbf{F}_{\text{fused}}$ with residual connection by (13)
return: $\mathbf{F}_{\text{fused}}$;

C. Spatial-Semantic-Guided Cross-Modal Fusion

Existing cross-modal fusion methods typically focus on feature affinity, often overlooking the comprehensive exploration of spatial distributions and semantic relationships among land covers. This oversight frequently compromises the semantic consistency of the fused representations. To overcome this limitation, we propose the SGCF module, which integrates spatial coordinates and semantic distribution priors to generate a dynamically adaptive bias matrix. The injection of this matrix into the attention computation enforces the alignment of fused features with the intrinsic scene distribution, enhancing semantic consistency. The computational procedure for the SGCF module is indicated in Algorithm 1.

To establish effective relational constraints, the features $\mathbf{F}_{\text{opt}}$ and $\mathbf{F}_{\text{geo}}$ are flattened into token sequences $\mathbf{T}_{\text{opt}}$ and $\mathbf{T}_{\text{geo}}$, respectively. On the one hand, to capture potential categorical dependencies, the features are projected into a categorical semantic space to estimate the semantic distribution probability for each modality

$$\mathbf{P}_m = \text{Softmax}\left(\mathbf{T}_m\mathbf{W}_{\text{cls}}\right) \quad (8)$$

where $m \in \{\text{opt}, \text{geo}\}$, $\mathbf{W}_{\text{cls}}$ is a learnable linear projection shared by both modalities to map optical and geometric tokens into the same category semantic space. On the other hand, to

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encode the spatial arrangement, an independent normalized 2-D coordinate grid $\mathbf{G} \in \mathbb{R}^{N\times 2}$ is constructed. Specifically, for a feature map with a spatial resolution of $H \times W$, the coordinate vector of the nth token located at (h, w) is defined as

$$\mathbf{G}_n = \left[\frac{2h}{H-1} - 1, \frac{2w}{W-1} - 1\right]. \tag{9}$$

This normalization ensures that spatial coordinates are strictly restricted within the range of $[-1, 1]$, providing a unified metric baseline for subsequent calculations of relative geometric relationships. Subsequently, a high-dimensional logical embedding space is constructed to dynamically evaluate relationships. By fusing the spatial grid $\mathbf{G}$ and the semantic prior $\mathbf{P}_m$ via a multilayer perceptron (MLP), the spatial-semantic embedding is generated

$$\mathbf{E}_m = \mathrm{MLP}\left(\mathrm{Concat}\left(\mathbf{G}, \mathbf{P}_m\right)\right). \tag{10}$$

In this spatial-semantic space, the direction and magnitude of the feature vectors implicitly encode the structural rules of the scene. The bias matrix $\mathbf{B}_{\text{guide}}$ is computed to model the pairwise logical compatibility between multimodal features

$$\mathbf{B}_{\text{guide}} = \tanh\left(\mathbf{E}_{\text{opt}}\mathbf{E}_{\text{geo}}^{\mathrm{T}}\right). \tag{11}$$

The bias matrix $\mathbf{B}_{\text{guide}}$ is constrained within the interval $[-1, 1]$ by the tanh function, where positive and negative values serve as promotion and suppression signals, respectively. The former aims to enhance interactions in semantically compatible regions, while the latter is used to attenuate associations in logically conflicting areas. Subsequently, $\mathbf{B}_{\text{guide}}$ is injected into the cross-modal attention as a spatial-semantic prior. In the rectified cross-attention calculation, $\mathbf{Q}_{\text{opt}}$ is generated from $\mathbf{F}_{\text{opt}}$ to maintain semantic dominance, while $\mathbf{K}_{\text{geo}}$ and $\mathbf{V}_{\text{geo}}$ are generated from $\mathbf{F}_{\text{geo}}$ to inject structural guidance

$$\mathbf{F}_{\text{att}} = \mathrm{Softmax}\left(\frac{\mathbf{Q}_{\text{opt}}\mathbf{K}_{\text{geo}}^{\top}}{\sqrt{d_k}} + \mathbf{B}_{\text{guide}}\right)\mathbf{V}_{\text{geo}}. \tag{12}$$

By incorporating $\mathbf{B}_{\text{guide}}$ prior to the Softmax operation, the attention distribution is effectively guided using prior logic. Finally, the attention output is combined with $\mathbf{F}_{\text{opt}}$ via a residual connection and processed by a feed-forward network (FFN) to obtain the multiscale features $\mathbf{F}_{\text{fused}}$

$$\mathbf{F}_{\text{fused}} = \mathrm{FFN}\left(\mathrm{LN}\left(\mathbf{F}_{\text{att}} + \mathbf{F}_{\text{opt}}\right)\right). \tag{13}$$

The SGCF module explicitly models spatial-semantic priors as a bias matrix that guides cross-modal feature fusion in accordance with land cover coexistence and exclusivity patterns, enhancing both semantic consistency and class discriminability of the fused features.

D. PSR Decoder

The SGCF module generates robust fused features $\mathbf{F}_{\text{fused}}$ by rectifying spatial-semantic relationships. To bridge the scale discrepancy between deep semantic features and shallow textural details, we design the PSR decoder as the final fusion and refinement stage to perform structural recovery and boundary refinement of $\mathbf{F}_{\text{fused}}$. The semantic constraints introduced by SGCF are preserved during decoding and further reflected in improved boundary continuity and structural integrity.

As shown in Fig. 5(a), in the progressive multiscale fusion stage, features from different resolutions are first bidirectionally aligned through dense upsampling and downsampling, then concatenated and projected to enable cross-scale interaction. This process enables progressive interaction between deep semantic information and shallow spatial details across different scales. Subsequently, to unify the output space, all low-resolution features are upsampled to the spatial resolution of the highest-resolution feature $\mathbf{F}_{\text{fused}}^{1}$. The aligned features are then concatenated along the channel dimension and projected by a 1×1 convolution to generate the unified representation $\mathbf{F}_{\text{uni}}$

$$\mathbf{F}_{\text{uni}} = \mathrm{Conv}_{1\times 1}\left(\mathrm{Concat}\left(\mathbf{F}_{\text{fused}}^{1}, \mathrm{Up}\left(\left\{\mathbf{F}_{\text{fused}}^{i}\right\}_{i=2}^{4}\right)\right)\right). \tag{14}$$

To further refine structural features, we introduce the multidirectional structure refinement block, as shown in Fig. 5(b). This block draws inspiration from the state space model (SSM) [36]. SSM models visual stimuli as a linear time-varying system, establishing a mapping between the 1-D input $x(t)$ and output $y(t)$ via a hidden state $h(t)$

$$\begin{aligned} \mathbf{h}'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) + \mathbf{D}x(t) \end{aligned} \tag{15}$$

where $\mathbf{A}$ is the evolution parameter, $\mathbf{B}$ and $\mathbf{C}$ are projection parameters, and $\mathbf{D}$ denotes the direct feedthrough parameter. To enable efficient computation, we employ the zero-order hold method, introducing a time-scale parameter Δ to discretize the aforementioned continuous dynamic system

$$\begin{aligned} \bar{\mathbf{A}} &= \exp\left(\Delta\mathbf{A}\right) \\ \bar{\mathbf{B}} &= \left(\Delta\mathbf{A}\right)^{-1}\left(\exp\left(\Delta\mathbf{A}\right) - \mathbf{I}\right)\Delta\mathbf{B} \end{aligned} \tag{16}$$

where $\mathbf{I}$ denotes the identity matrix. Distinct from traditional linear time-invariant models, to implement the selective scan mechanism, the parameters Δ, $\bar{\mathbf{B}}$, and $\mathbf{C}$ are designed to be dependent on the input x_t. This input dependence empowers the model to selectively propagate or forget information based on the current context. Consequently, the derived discrete recursive form is expressed as

$$\begin{aligned} \mathbf{h}_t &= \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t \\ y_t &= \mathbf{C}h_t + \mathbf{D}x_t. \end{aligned} \tag{17}$$

Although 2-D selective scan (SS2D) [37] extends SSMs to the image domain via directional sequence flattening, it exhibits certain limitations in spatial feature modeling. Specifically, restricting the scanning paths to four cardinal directions introduces an inherent Manhattan metric bias [55]. This limits the model's ability to capture continuous nonaxial features with isotropy. To overcome this limitation, we propose an eight-directional scanning mechanism that expands the scan path set to D_8, which includes horizontal, vertical, principal diagonal, and auxiliary diagonal directions. This comprehensive directional coverage establishes an isotropic receptive field with improved geometric uniformity. The outputs from the eight scanning directions are remapped back to the original 2-D grid through inverse directional indices, ensuring that responses from different scanning paths are spatially realigned

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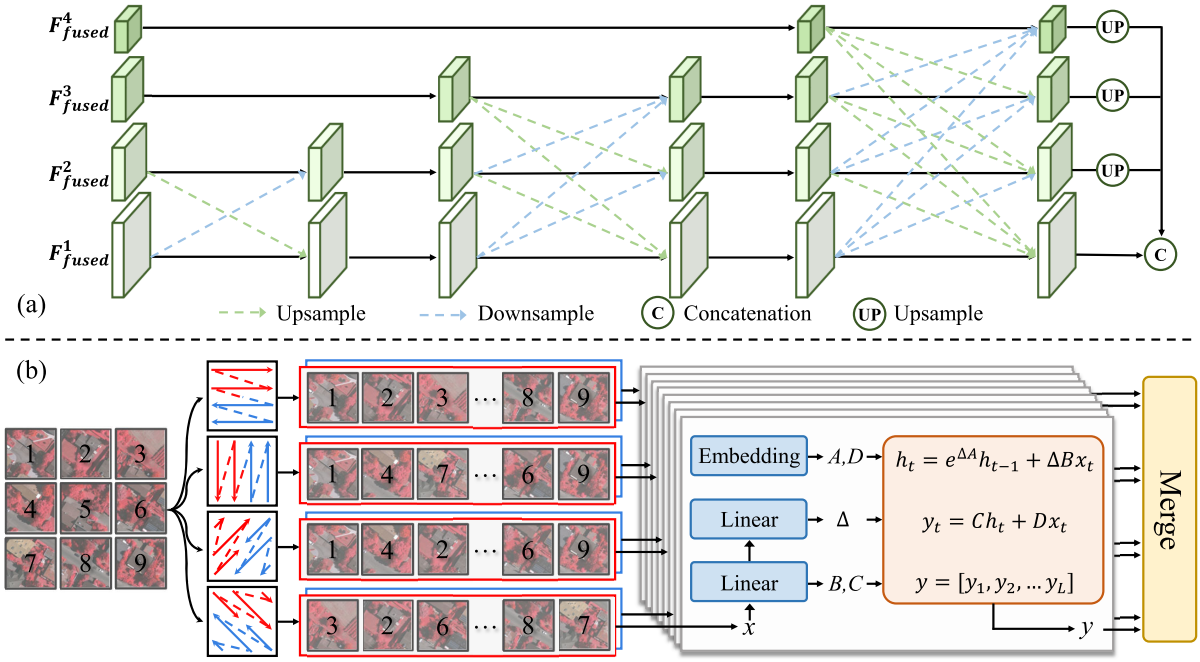


Fig. 5. Two key components of the PSR decoder. (a) Progressive multiscale fusion structure. (b) Multidirectional structure refinement block.

at each pixel position. The aligned features are then aggregated and normalized to produce the final refined feature

$$\mathbf{F}_{\text{refined}} = \text{LN}\left(\sum_{d \in D_8} \text{R}_d\left(\text{Scan}\left(\mathbf{F}_{\text{uni}}, d\right)\right)\right) \tag{18}$$

where $\text{Scan}(\cdot, d)$ denotes the selective scan operation along direction d, and R_d denotes the inverse directional reordering operation that maps the scanned sequence back to the original 2-D spatial grid. The realigned directional features are then aggregated across all scanning paths and normalized to form the refined feature $\mathbf{F}_{\text{refined}}$. Since PSR operates on the SGCF-enhanced fused features, the directional scanning process refines spatial structures while maintaining the semantic relationship constraints established during cross-modal fusion. Finally, $\mathbf{F}_{\text{refined}}$ is fed into the segmentation head and mapped to the category space to generate the final prediction.

E. Loss Function

A joint loss function is designed to supervise the training process while ensuring the robustness and effective utilization of the semantic prior $\mathbf{P}_m$. The total loss function $\mathcal{L}_{total}$ consists of the loss $\mathcal{L}_{seg}$, which is widely used in segmentation tasks, and the negative entropy regularization loss $\mathcal{L}_{\mathbf{P}_m}$. The loss $\mathcal{L}_{seg}$ includes the cross-entropy loss $\mathcal{L}_{ce}$ and the dice loss $\mathcal{L}_{dice}$. $\mathcal{L}_{ce}$ is used to quantify the discrepancy between the prediction and the ground truth, while $\mathcal{L}_{dice}$ is used to mitigate the impact of class imbalance

$$\mathcal{L}_{\text{seg}} = \mathcal{L}_{\text{ce}} + \mathcal{L}_{\text{dice}}. \tag{19}$$

To mitigate the risk of $\mathbf{P}_m$ being overconfident in potentially erroneous predictions and to reinforce the role of the prior distribution, an entropy regularization loss is introduced

$$\mathcal{L}_{\mathbf{P}_m} = \lambda \sum_{x \in \Omega} \sum_{c=1}^{C} \left[\mathbf{P}_{m,c}(x) \cdot \log\left(\mathbf{P}_{m,c}(x)\right)\right] \tag{20}$$

where Ω denotes the set of all pixels, C is the total number of categories, and $\mathbf{P_{m,c}}(x)$ is the predicted probability for class c at pixel x from P_m. λ is a hyperparameter that controls the strength of the contribution of the entropy regularization term within the total loss function, adjusting the degree of the entropy constraint on $\mathbf{P}_m$. This aims to ensure that $\mathbf{P}_m$ maintains a relatively high entropy, providing an accurate and flexible prior for the cross-modal fusion. The total loss function is defined as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}} + \mathcal{L}_{\mathbf{P}_m}. \tag{21}$$

This joint loss enables $\mathbf{P}_m$ to provide reliable semantic priors for fusion. Its potential inaccuracies are corrected by Lseg and explicitly constrained by entropy regularization, effectively improving training performance and robustness.

IV. Experiments and Analysis

A. Dataset

For the experiments, we selected four representative multimodal RSI semantic segmentation datasets: ISPRS Vaihingen, ISPRS Potsdam, DDHR, and WHU-OPT-SAR. These datasets cover multimodal combinations of optical-DSM and optical-SAR, encompassing typical land covers such as buildings, roads, farmland, and vegetation. Fig. 6 presents sample images from the four datasets.

1) ISPRS Vaihingen: The ISPRS Vaihingen dataset contains 33 high-resolution orthophoto images with an average size of 2494 × 2064 pixels and a ground sampling distance of 9 cm. The official ISPRS 2-D Semantic Labeling Benchmark

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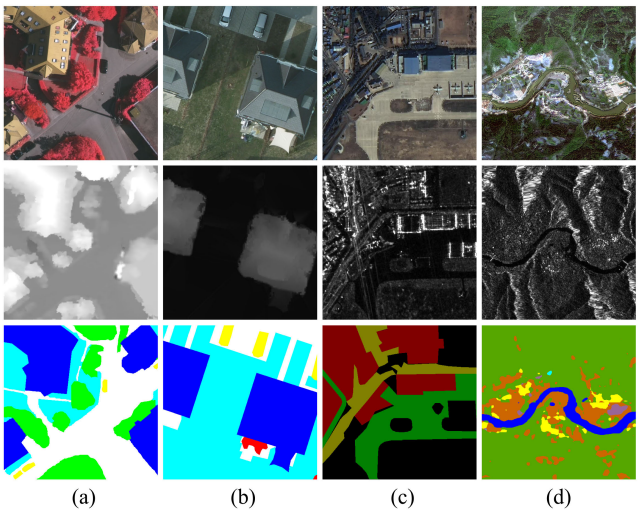


Fig. 6. Data samples from the four benchmark datasets used in our experiments. (a) ISPRS Vaihingen dataset. (b) ISPRS Potsdam dataset. (c) DDHR dataset. (d) WHU-OPT-SAR dataset.

provides corresponding DSM data and annotations for each optical image, covering six semantic categories: impervious surfaces, building, low vegetation, tree, car, and clutter. In the data preprocessing stage, we first divided the original large-scale images. Twenty-three images were used for training, three images for validation, and seven images for testing. To meet deep learning input requirements, the divided large-scale images were cropped into 512×512 pixel patches using a sliding-window strategy. After cropping, 1215 training samples, 174 validation samples, and 347 test samples were obtained.

2) ISPRS Potsdam: The ISPRS Potsdam dataset contains 38 high-resolution images, with each image having a size of 6000×6000 pixels and a spatial resolution of 5 cm. The official ISPRS 2-D Semantic Labeling Benchmark provides corresponding DSM data and annotations for each optical image, using the same semantic categories as the Vaihingen dataset. Consistent with the Vaihingen dataset, we first divided the original large-scale images, with 27 images used for training, four images for validation, and four images for testing. Then, the divided large-scale images were cropped into 512×512 pixel patches using a sliding-window strategy. After cropping, 3134 training samples, 448 validation samples, and 895 test samples were obtained.

3) DDHR: The DDHR dataset is a high-resolution multimodal dataset composed of Gaofen-2 optical images and Gaofen-3 SAR images. The dataset unifies the spatial resolution to 1 m and covers three typical regions: Xi'an, Dongying, and Pohang. The annotations for this dataset cover five semantic categories: building, water, farmland, greenery, and road. In the data preprocessing stage, the optical and SAR images were kept spatially aligned, and a consistent sliding-window strategy was adopted to crop them into 2550 image patches of 512×512 pixels. The divided samples include 1785 training samples, 255 validation samples, and 510 test samples.

4) WHU-OPT-Synthetic Aperture Radar: The WHU-OPT-SAR dataset was released by Wuhan University. It contains 100 pairs of well-registered Gaofen-1 optical images and Gaofen-3 SAR images, with each image having a resolution of 5556×3704 pixels. The annotations cover eight semantic categories: farmland, city, village, water, forest, road, others, and background. In the data preprocessing stage, we first divided the original images, with 70 images used for training, ten images for validation, and 20 images for testing. Then, the divided images were cropped into 512×512 pixel patches using a sliding-window strategy. After cropping, 4900 training samples, 700 validation samples, and 1400 test samples were obtained.

B. Implementation Details

All experiments were implemented using the PyTorch framework and conducted on NVIDIA GeForce RTX 3090 Ti GPUs with 24 GB of memory for training and inference. A consistent experimental setup was adopted across all models. For parameter optimization, we used the AdamW optimizer with an initial learning rate of 3.5×10^{-4} and a weight decay of 1×10^{-3}. The learning rate was dynamically adjusted using a cosine annealing schedule to improve convergence stability. All models were trained for 100 epochs with a fixed batch size of 4, and the model checkpoint from the final training epoch was used for testing. To ensure a fair evaluation of network performance, no additional data augmentation was applied after the sliding-window cropping process. To improve the statistical reliability and stability of the experimental results, all methods were independently trained and evaluated three times under the same fixed data split, with different seeds used for each run. The mean and standard deviation are reported.

C. Evaluation Metrics

To comprehensively evaluate the performance of R2SNet in the RSI semantic segmentation task, we utilized comprehensive evaluation metrics including overall accuracy (OA), mean intersection over union (mIoU), boundary intersection over union (BIoU), and mean $F1$ score (m$F1$). Additionally, to measure computational complexity and resource consumption, we also report floating-point operations (FLOPs) and parameters (Params). OA intuitively represents the pixel-level classification accuracy of the model, defined as

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{22}$$

where TP, TN, FP, and FN represent the number of true positive, true negative, false positive, and false negative pixels, respectively. mIoU serves as a core metric for evaluating segmentation tasks, designed to measure the overlap between predictions and labels. The calculation is as follows:

$$\mathrm{mIoU} = \frac{1}{N} \sum_{i=1}^{N} \frac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i} \tag{23}$$

where N denotes the total number of categories, and i represents the category index. BIoU is utilized to quantify the overlap between the predicted boundary and the ground-truth boundary. The boundary region is generated by dilating the extracted boundary with a radius of 5 pixels

$$\mathrm{BIoU} = \frac{\left|\mathrm{B}\left(\mathrm{S}_{gt}\right) \cap \mathrm{B}\left(\mathrm{S}_{p}\right)\right|}{\left|\mathrm{B}\left(\mathrm{S}_{gt}\right) \cup \mathrm{B}\left(\mathrm{S}_{p}\right)\right|} \tag{24}$$

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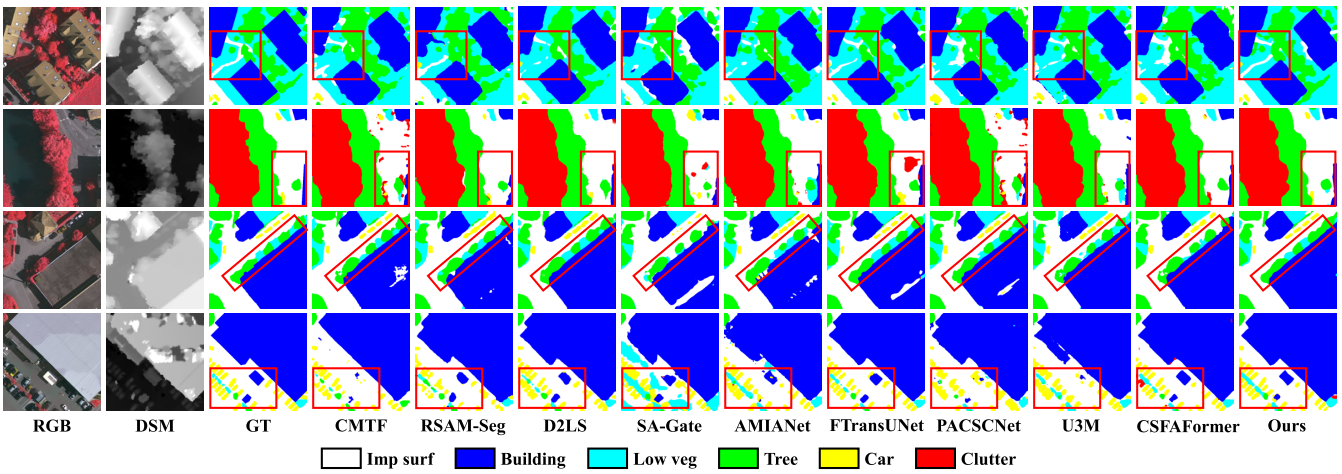


Fig. 7. Visualization results of different methods on the Vaihingen dataset. Red boxes highlight regions with significant differences.

TABLE I

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE VAIHINGEN DATASET. BOLD VALUES INDICATE THE BEST RESULTS, AND UNDERLINED VALUES INDICATE THE SECOND-BEST RESULTS

Type	Method	Venue	IoU (%) Per Category					OA (%)	mF1 (%)	BIoU (%)	mIoU (%)
			Imp.	Bui.	Low.	Tree	Car				
Unimodal	A2FPN [28]	CVPR'21	86.90±0.32	91.42±0.46	78.77±0.33	84.30±0.23	72.05±0.17	92.03±0.15	88.03±0.27	30.46±0.19	79.26±0.44
	CMTFNet [35]	TGRS'23	86.93±0.17	91.17±0.50	78.59±0.34	84.86±0.28	69.96±0.90	92.12±0.17	88.26±0.17	32.04±0.13	79.59±0.19
	RS3Mamba [38]	GRSL'24	87.43±0.05	91.25±0.09	78.85±0.25	84.64±0.14	60.98±0.12	92.40±0.04	89.06±0.39	34.44±0.20	81.35±0.58
	RSAM-Seg [45]	RS'25	87.29±0.20	91.38±0.08	78.51±0.57	84.51±0.07	59.66±1.30	92.18±0.08	88.38±0.15	32.64±0.05	80.31±0.18
	D2LS [56]	ICCV'25	88.89±0.28	93.78±0.19	81.27±0.76	84.92±0.47	71.87±0.94	93.06±0.23	90.62±0.25	38.26±0.20	84.19±0.40
Multimodal	SAGate [51]	ECCV'20	84.27±0.64	90.33±1.06	78.42±0.37	81.21±1.51	64.17±2.25	90.31±0.41	87.39±0.95	35.97±1.18	78.16±0.59
	CMX [57]	TITS'24	91.31±0.33	94.55±0.38	83.73±0.18	87.33±0.13	74.14±0.27	93.62±0.13	92.78±0.13	50.31±0.50	86.77±0.22
	PACSCNet [14]	ESWA'24	91.46±0.10	94.60±0.05	84.15±0.22	88.04±0.18	76.46±0.15	93.75±0.06	93.18±0.05	47.50±0.45	87.41±0.10
	AMIANet [21]	TGRS'24	92.01±0.08	94.69±0.01	84.23±0.03	88.14±0.03	79.94±0.48	94.22±0.02	93.45±0.11	50.58±0.33	87.81±0.20
	FTransUNet [19]	TGRS'24	91.12±0.65	93.25±0.18	83.04±0.18	87.46±0.24	73.51±0.19	93.41±0.23	92.25±0.50	45.48±0.58	86.06±0.34
	MFNet [58]	TGRS'25	89.31±0.20	93.09±0.04	82.60±0.21	85.92±0.10	75.70±0.65	93.14±0.09	91.63±0.40	42.07±0.12	85.27±0.53
	StitchFusion [20]	MM'25	90.76±0.27	93.13±0.19	82.76±0.68	86.19±0.54	75.37±0.97	93.18±0.22	91.75±0.22	44.22±0.98	85.84±0.36
	U3M [53]	PR'25	91.16±0.71	93.42±0.35	82.79±0.82	85.34±0.37	73.56±2.21	93.39±0.35	92.04±0.50	42.28±1.07	86.05±0.74
	CSFAFormer [52]	IF'26	92.11±0.07	94.29±0.14	84.38±0.22	88.19±0.20	78.46±0.13	94.66±0.13	93.56±0.06	51.09±0.32	88.05±0.16
	R2SNet (Ours)	-	**92.68±0.24**	**95.27±0.31**	**84.69±0.36**	**88.85±0.51**	**83.50±0.03**	**94.91±0.20**	**94.16±0.18**	**53.41±0.94**	**89.06±0.33**

where $B(S_{gt})$ denotes the ground-truth boundary and $B(S_p)$ denotes the predicted boundary. The $F1$ score, as the harmonic mean of precision and recall, evaluates both the accuracy and completeness of predictions. m$F1$ is the average of all category $F1$ scores. The m$F1$ is calculated as follows:

$$\text{mF1} = \frac{1}{N}\sum_{i=1}^{N}\frac{2\cdot \text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i}. \tag{25}$$

D. Comparison With State-of-the-Art Methods

The performance of the proposed R2SNet was comprehensively evaluated by comparing it with the 14 advanced methods developed in recent years. Five of these methods used optical RSI alone. These unimodal methods quantified the contribution of auxiliary modalities, thereby highlighting the advantages of multimodal fusion over unimodal methods. The remaining nine methods utilized multimodal RSI.

1) Comparison on the ISPRS Vaihingen Dataset: The quantitative results on the Vaihingen dataset are illustrated in Table I, where R2SNet performed superiorly across all evaluation metrics. Specifically, R2SNet achieved the OA of 94.91% ± 0.20%, m$F1$ of 94.16% ± 0.18%, BIoU of 53.41% ± 0.94%, and mIoU of 89.06% ± 0.33%. Compared to the second-best method, CSFAFormer, our method achieved an improvement of 1.01% in mIoU. Particularly for the challenging car class, R2SNet achieved an IoU of 83.50% ± 0.03%, outperforming the second-best method AMIANet by 3.56%. This observable improvement underscores the favorable capability of the model in accurately capturing small objects and fine-grained details. Fig. 7 provides a visual comparison of the segmentation results on the Vaihingen dataset, with red boxes highlighting regions where the performance differences between methods are pronounced. In general, the segmentation results generated by R2SNet align more closely with ground truth and indicate notable advantages in several critical aspects. Specifically, for the densely distributed cars, comparative methods, including CSFAFormer and U3M, frequently struggle to completely distinguish individual instances. R2SNet accurately identifies independent

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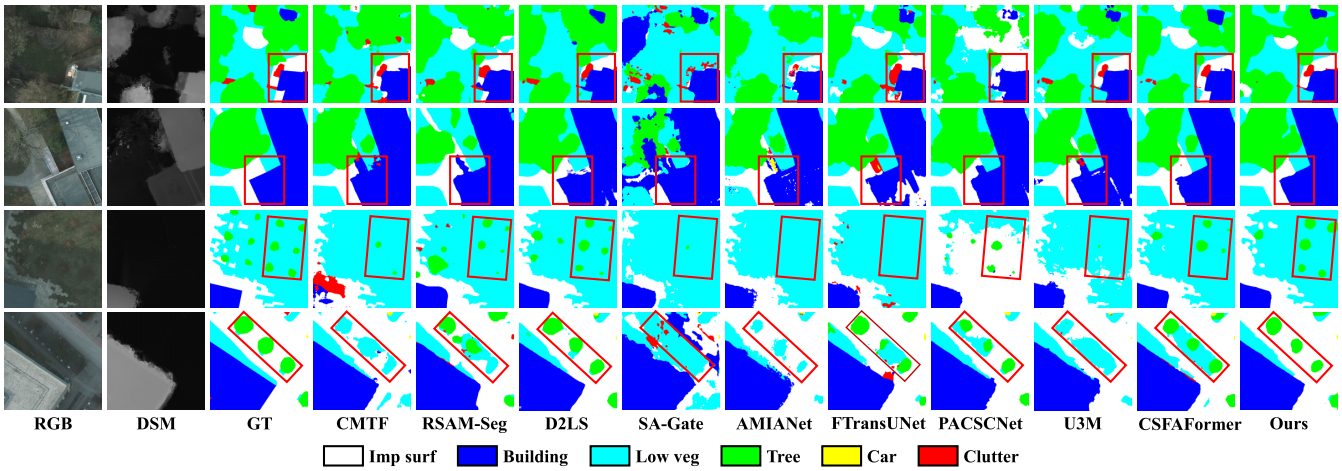


Fig. 8. Visualization results of different methods on the Potsdam dataset. Red boxes highlight regions with significant differences.

TABLE II

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE POTSDAM DATASET. BOLD VALUES INDICATE THE BEST RESULTS, AND UNDERLINED VALUES INDICATE THE SECOND-BEST RESULTS

Type	Method	Venue	IoU (%) Per Category					OA (%)	mF1 (%)	BIoU (%)	mIoU (%)
			Imp.	Bui.	Low.	Tree	Car				
Unimodal	A2FPN [28]	CVPR'21	78.45±0.39	87.20±0.25	68.91±0.30	62.91±0.87	75.75±0.61	85.89±0.17	81.44±0.36	21.64±0.30	69.42±0.42
	CMTFNet [35]	TGRS'23	78.50±0.59	87.03±0.61	68.33±0.45	64.28±0.62	75.52±0.77	85.31±0.42	80.84±0.66	22.47±0.27	69.07±0.80
	RS3Mamba [38]	GRSL'24	77.32±0.32	86.76±0.53	67.26±0.64	62.56±0.30	73.83±0.57	84.78±0.49	80.35±0.55	21.65±0.17	68.42±0.59
	RSAM-Seg [45]	RS'25	80.47±0.54	88.36±0.76	70.74±0.52	64.99±0.77	78.06±0.81	79.65±0.49	74.19±0.60	21.19±0.20	70.59±0.75
	D2LS [56]	ICCV'25	82.97±0.19	89.98±0.30	73.54±0.29	72.18±0.32	79.04±0.25	88.47±0.11	84.67±0.10	25.30±0.34	74.25±0.13
Multimodal	SAGate [51]	ECCV'20	77.87±0.90	86.51±0.58	66.19±0.72	61.11±0.79	70.98±0.99	85.19±0.13	80.66±0.32	20.14±0.11	68.91±0.24
	CMX [57]	TITS'24	78.52±0.48	87.40±0.32	68.39±0.44	64.64±0.53	74.68±0.71	85.93±0.10	81.63±0.26	22.10±0.18	69.65±0.19
	PACSCNet [14]	ESWA'24	83.44±0.20	92.19±0.11	73.57±0.22	72.27±0.21	81.80±0.43	89.14±0.05	85.25±0.11	28.09±0.16	75.17±0.16
	AMIANet [21]	TGRS'24	82.66±0.21	91.56±0.10	72.62±0.14	70.71±0.09	81.14±0.26	88.01±0.07	84.14±0.20	26.79±0.06	73.98±0.21
	FTransUNet [19]	TGRS'24	82.80±0.44	91.38±0.16	73.51±0.46	71.95±0.30	80.26±0.24	88.52±0.12	84.62±0.45	26.39±0.20	74.11±0.24
	MFNet [58]	TGRS'25	81.75±0.25	90.72±0.16	72.39±0.18	70.56±0.23	79.11±0.28	87.91±0.12	83.27±0.37	23.95±0.09	72.96±0.37
	StitchFusion [20]	MM'25	82.11±0.70	91.69±0.13	72.11±0.87	70.63±0.74	80.46±0.80	88.29±0.45	83.51±0.19	25.04±0.31	73.59±0.58
	U3M [53]	PR'25	82.54±0.73	91.72±0.42	72.50±0.27	70.60±0.75	81.78±0.29	88.45±0.29	84.26±0.17	26.68±0.03	74.05±0.25
	CSFAFormer [52]	IF'26	83.75±0.33	**92.60±0.31**	73.57±0.32	72.60±0.22	81.96±0.16	89.19±0.25	85.50±0.30	28.02±0.09	75.96±0.37
	R2SNet (Ours)	-	**84.51±0.16**	92.37±0.21	**74.45±0.23**	**73.49±0.41**	**83.05±0.09**	**89.36±0.07**	**86.44±0.08**	**30.73±0.16**	**76.98±0.11**

vehicles, showcasing exceptional small object capture and detail preservation capabilities. Furthermore, observing the transition areas between buildings and impervious surfaces in rows two and three reveals that models such as SAGate and AMIANet tend to produce jagged edges or erroneous class predictions, leading to blurred boundaries. In contrast, R2SNet generates smooth, well-defined building boundaries that effectively preserve the original geometric features of land cover, demonstrating better boundary integrity. In vegetation areas, many methods suffer from noisy or semantically incomplete segmentation results due to severe interference from shadow occlusions or textural similarities. R2SNet exhibits greater robustness, achieving consistent and coherent predictions.

2) Comparison on the ISPRS Potsdam Dataset: Table II presents the quantitative results of R2SNet on the Potsdam dataset. Our method achieves state-of-the-art performance, with an OA of 89.36% ± 0.07%, m$F1$ of 86.44% ± 0.08%, BIoU of 30.73% ± 0.16%, and mIoU of 76.98% ± 0.11%. For the low vegetation and tree categories, which are prone to spectral confusion, R2SNet indicates strong discriminative ability, achieving IoUs of 74.45% ± 0.23% and 73.49 % ± 0.41%, respectively. In particular, the low vegetation and tree categories show improvements of 0.88% and 0.89% over the second-best results, respectively. The results indicate that R2SNet effectively leverages geometric priors to distinguish land covers with similar textures but differing elevations, addressing the limitations of spectral features alone and yielding robust segmentation performance. As shown in Fig. 8, R2SNet performs well in parsing complex urban scenes. The Potsdam dataset is renowned for its dense historical building complexes and narrow street networks, which pose a significant challenge to the fine-grained parsing capabilities of the methods. When confronted with continuous buildings and intricate shadow occlusions, comparative methods such as U3M and AMIANet frequently experience semantic fragmentation, occasionally even misclassifying streets as impervious surfaces. R2SNet maintains high semantic coherence within objects, producing more complete and consistent segmentations. Notably, the PSR decoder's structure refinement contributes to more accurate delineation of building

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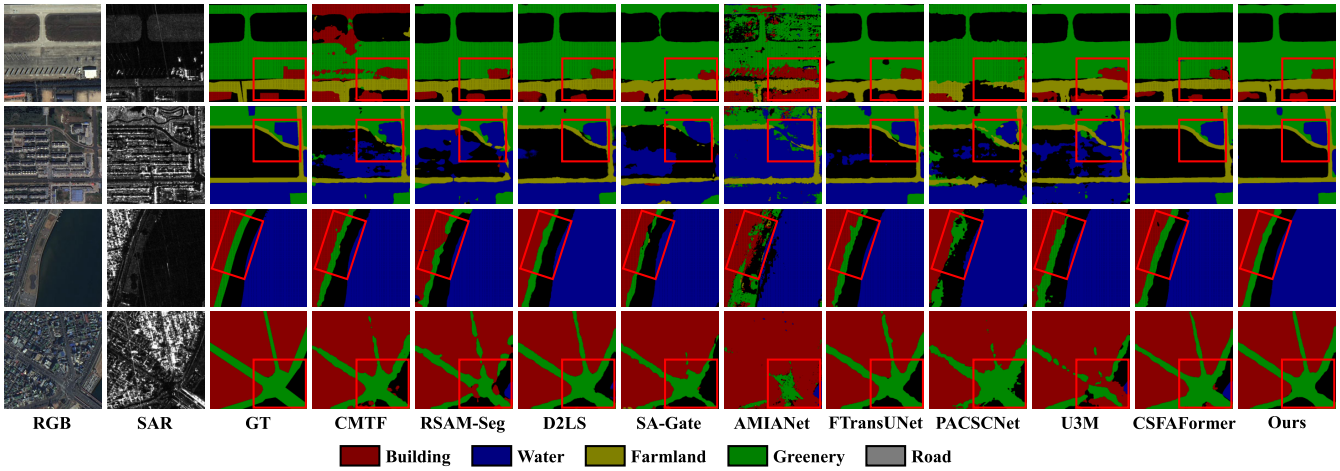


Fig. 9. Visualization results of different methods on the DDHR dataset. Red boxes highlight regions with significant differences.

TABLE III

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE DDHR DATASET. BOLD VALUES INDICATE THE BEST RESULTS, AND UNDERLINED VALUES INDICATE THE SECOND-BEST RESULTS

Type	Method	Venue	IoU (%) Per Category					OA (%)	mF1 (%)	BIoU (%)	mIoU (%)
			Bui.	Wat.	Far.	Gre.	Road				
Unimodal	A2FPN [28]	CVPR'21	94.65±0.16	94.01±0.19	88.76±0.14	84.80±0.53	69.36±0.38	95.17±0.12	92.38±0.15	32.27±0.55	86.32±0.24
	CMTFNet [35]	TGRS'23	93.17±0.39	93.23±0.20	87.15±0.27	81.28±0.84	66.39±0.93	94.24±0.20	91.11±0.31	28.54±0.25	84.25±0.50
	RS3Mamba [38]	GRSL'24	94.35±0.19	94.34±0.17	89.73±0.09	86.12±0.29	72.81±0.18	95.46±0.08	93.11±0.07	37.19±0.56	87.47±0.12
	RSAM-Seg [45]	RS'25	92.55±0.48	92.93±0.09	86.71±0.30	79.49±0.73	66.15±1.03	93.90±0.26	90.72±0.41	27.67±0.36	83.97±0.59
	D2LS [56]	ICCV'25	94.95±0.20	94.38±0.21	89.58±0.19	84.59±0.36	67.15±0.94	95.30±0.09	92.20±0.25	33.15±0.55	86.13±0.38
Multimodal	SAGate [51]	ECCV'20	94.36±0.05	92.99±0.24	87.02±0.53	82.35±0.58	67.82±1.14	94.52±0.16	91.53±0.29	28.25±0.74	84.91±0.46
	CMX [57]	TITS'24	91.38±0.68	93.46±0.42	87.03±0.64	77.24±1.09	63.50±0.73	93.47±0.53	90.00±0.66	29.11±0.97	82.53±0.65
	PACSCNet [14]	ESWA'24	95.64±0.10	95.11±0.18	91.70±0.25	89.38±0.44	75.83±0.69	96.18±0.16	94.47±0.07	43.87±0.50	89.37±0.14
	AMIANet [21]	TGRS'24	94.36±0.59	93.95±0.53	88.72±0.72	84.04±0.95	68.79±0.89	95.03±0.65	92.16±0.81	31.99±1.03	85.98±0.88
	FTransUNet [19]	TGRS'24	95.68±0.46	95.76±0.36	91.73±0.38	89.09±0.68	75.66±0.68	96.21±0.18	94.56±0.27	43.64±0.77	89.49±0.43
	MFNet [58]	TGRS'25	92.51±0.43	92.25±0.16	84.86±0.03	76.42±0.19	62.67±0.40	94.77±0.16	92.14±0.02	29.53±0.08	84.20±0.07
	StitchFusion [20]	MM'25	93.08±0.64	92.73±0.59	86.99±0.47	81.74±0.66	67.71±0.91	94.65±0.49	91.95±0.86	28.07±0.27	84.65±0.64
	U3M [53]	PR'25	94.68±0.81	93.42±0.42	87.73±0.58	81.64±0.83	68.04±1.04	94.68±0.58	92.40±0.77	31.14±0.49	85.90±0.73
	CSFAFormer [52]	IF'26	93.63±0.12	94.18±0.19	90.32±0.22	86.89±0.17	72.41±0.44	95.63±0.05	92.57±0.09	41.88±0.31	87.18±0.16
	R2SNet (Ours)	-	**96.30±0.20**	**95.85±0.33**	**92.54±0.40**	**89.40±0.61**	**75.86±0.63**	**96.79±0.19**	**94.80±0.36**	**45.16±0.74**	**90.11±0.53**

boundaries, mitigating boundary adhesion and blur that commonly afflict traditional methods and yielding predictions with refined boundaries and intact structures. The areas marked by red boxes, where significant differences are observed, are typically located in densely building-filled and shaded boundary regions, highlighting R2SNet's strength in capturing fine-grained spatial structures. These qualitative results are consistent with the quantitative improvements reported in Table II, supporting the conclusion that combining geometric prior with incremental structural refinement enhances the robustness of urban land cover classification.

3) Comparison on the DDHR Dataset: The quantitative results of different methods on the DDHR dataset are presented in Table III. R2SNet achieved an OA of 96.79% ± 0.19%, m$F1$ of 94.80% ± 0.36%, BIoU of 45.16% ± 0.74%, and mIoU of 90.11% ± 0.53%. Compared with the second-best method, FTransUNet, R2SNet achieves an improvement of 0.62% in mIoU. In the building, water, and farmland categories, R2SNet reached IoUs of 96.30% ± 0.20%, 95.85% ± 0.33%, and 92.54% ± 0.40%, respectively. This outstanding performance is mainly attributed to the ADB encoder, which effectively tailors feature extraction to the distinct characteristics of each modality. This design enables robust extraction of structural texture information from SAR imagery, which is crucial for accurately delineating road and farmland boundaries. It also suppresses speckle noise through multiscale interaction, achieving more precise land-cover discrimination. Visualization results on the DDHR dataset shown in Fig. 9 further validate the advantages of R2SNet in multimodal fusion. At the water-land interface and narrow road segmentation tasks, many methods are constrained by noise interference in SAR images, leading to fragmented linear features such as roads or salt-and-pepper-like artifacts in homogeneous regions. R2SNet yields coherent and clear segmentations with reduced boundary artifacts. Particularly when distinguishing between categories with extremely similar spectra but subtle textural differences, such as farmland and greenery, our method indicates higher logical consistency. R2SNet successfully minimizes interclass confusion, illustrating the effectiveness of

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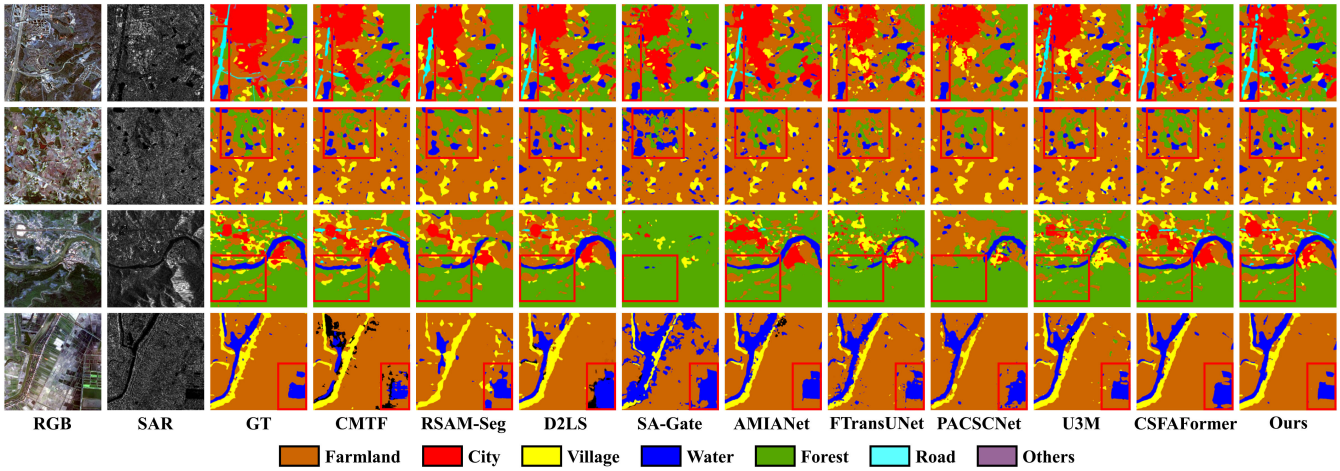


Fig. 10. Visualization results of different methods on the WHU-OPT-SAR dataset. Red boxes highlight regions with significant differences.

TABLE IV

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE WHU-OPT-SAR DATASET. BOLD VALUES INDICATE THE BEST RESULTS, AND UNDERLINED VALUES INDICATE THE SECOND-BEST RESULTS

Type	Method	IoU (%) Per Category						OA (%)	mF1 (%)	BIoU (%)	mIoU (%)
		For.	Vil.	Wat.	Far.	City	Road				
Unimodal	A2FPN [28]	88.66±0.08	75.47±0.04	69.42±0.32	59.75±0.49	64.39±0.35	19.52±1.33	87.20±0.05	57.36±0.36	19.64±0.32	47.87±0.24
	CMTFNet [35]	88.87±0.12	76.29±0.41	69.87±1.04	59.28±1.81	58.15±1.15	20.68±0.66	87.21±0.19	60.51±0.76	20.79±0.21	49.37±0.57
	RS3Mamba [38]	87.30±0.75	73.33±0.57	68.21±0.25	58.42±0.57	56.11±0.73	18.67±0.44	85.74±0.54	55.27±0.55	18.79±0.34	45.32±0.65
	RSAM-Seg [45]	85.56±0.25	70.00±0.23	62.15±0.44	45.66±1.00	58.63±1.68	18.84±0.86	83.71±0.22	52.92±0.54	17.85±0.34	44.34±0.14
	D2LS [56]	91.37±0.18	80.73±0.19	78.08±0.15	64.75±0.19	71.97±0.20	27.67±0.80	90.29±0.07	66.13±0.63	25.26±0.18	55.47±0.46
Multimodal	SAGate [51]	90.32±0.34	77.61±0.56	73.27±0.84	60.16±0.95	70.03±0.92	14.86±1.31	88.76±0.37	63.14±0.95	20.82±0.63	52.37±0.83
	CMX [57]	91.60±0.20	81.49±0.25	78.26±0.37	65.36±0.42	72.08±0.40	28.93±0.79	90.41±0.15	66.80±0.82	26.48±0.28	56.05±0.61
	PACSCNet [14]	90.36±0.14	80.63±0.19	78.47±0.26	61.05±0.48	70.46±0.49	28.83±0.37	89.46±0.09	63.52±0.77	24.26±0.31	53.48±0.23
	AMIANet [21]	92.54±0.10	82.97±0.13	79.46±0.49	67.75±0.58	73.86±0.60	39.91±0.28	91.36±0.06	68.56±0.91	28.22±0.20	58.23±0.89
	FTransUNet [19]	91.20±0.17	80.65±0.19	78.40±0.26	65.70±0.63	71.60±0.32	29.62±0.63	90.21±0.08	65.82±0.44	26.03±0.38	55.35±0.61
	MFNet [58]	89.89±0.15	78.77±0.15	75.97±0.26	59.97±0.49	69.04±0.40	29.55±0.20	89.05±0.02	59.51±0.51	21.86±0.19	50.70±0.27
	StitchFusion [20]	90.01±0.24	79.58±0.26	77.30±0.34	61.08±0.37	70.65±0.48	29.42±0.28	89.48±0.10	63.57±0.77	22.94±0.36	53.39±0.53
	U3M [53]	89.97±0.23	79.70±0.22	77.70±0.31	61.41±0.38	70.34±0.44	29.20±0.35	89.51±0.12	63.84±0.65	24.35±0.33	53.52±0.47
	CSFAFormer [52]	92.16±0.21	82.89±0.17	**81.90±0.22**	67.79±0.53	**73.87±0.18**	40.12±0.61	91.07±0.05	69.19±0.79	30.09±0.32	58.71±0.64
	R2SNet (Ours)	**92.66±0.09**	**83.75±0.23**	81.84±0.24	**68.62±0.70**	73.84±0.30	**41.56±0.39**	**91.78±0.07**	**70.11±0.81**	**30.60±0.37**	**59.54±0.59**

spatial-semantic priors in guiding cross-modal fusion and correcting erroneous predictions. Overall, these results support the robustness and generalizability of the proposed framework for multimodal DDHR segmentation under challenging imaging conditions.

4) Comparison on the WHU-OPT-SAR Dataset: Table IV illustrates that even in the more complex scenes of the WHU-OPT-SAR dataset, R2SNet maintained competitive performance, achieving an OA of 91.78% ± 0.07%, m$F1$ of 70.11% ± 0.81%, BIoU of 30.60% ± 0.37%, and mIoU of 59.54% ± 0.59%. Compared with the second-best method CSFAFormer, R2SNet slightly improves the mIoU by 0.83%. Although the numerical gain is moderate, R2SNet achieves the best overall mIoU and BIoU, indicating its advantage in balancing semantic classification accuracy and boundary quality in complex optical-SAR scenes. Despite the presence of numerous unstructured regions such as forests and villages within this dataset, our method attained the best IoUs of 92.66% ± 0.09%, 83.75% ± 0.23%, 68.62% ± 0.70%, and 41.56% ± 0.39% for forest, village, farmland, and road, respectively. In addition, R2SNet achieved competitive performance in the water and city categories, indicating its robustness across various types of land cover. As depicted in Fig. 10, the segmentation performance of different methods on the WHU-OPT-SAR dataset is visualized for comparison. Results indicate that R2SNet exhibits favorable capability in effectively handling transitional areas between large-scale land covers such as forests and farmland, while significantly reducing interclass confusion. Compared to methods that frequently produce fragmented results or misclassifications, particularly at complex boundaries like dense forests, winding rural roads, or the transition between villages and farmland, R2SNet consistently achieves more precise delineation. Our method leverages its multidirectional structure refinement block to adeptly capture subtle non-Manhattan structural features, which are crucial for distinguishing fine linear elements like narrow roads or precise water outlines. This enables R2SNet to successfully recover rich details, generating predictions that appear visually natural and align with geospatial distribution patterns.

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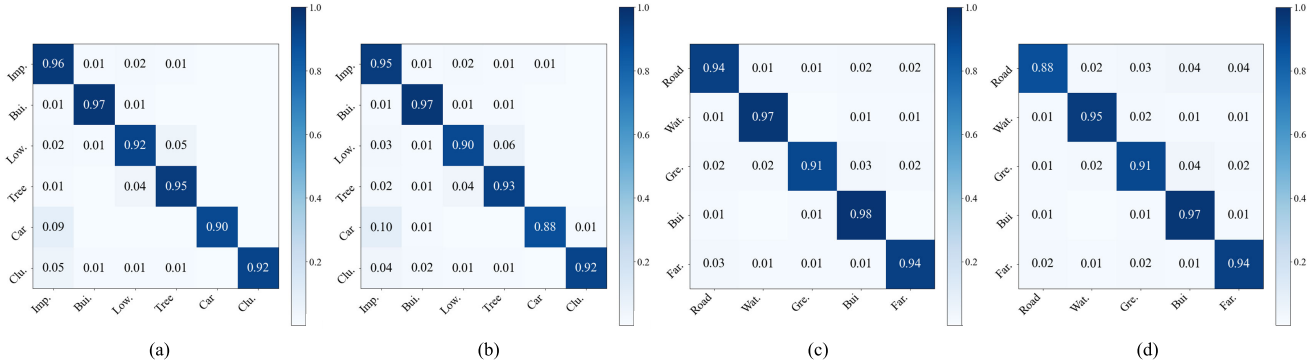


Fig. 11. Confusion matrices for modal ablation on the Vaihingen and DDHR datasets. (a) Multimodal Vaihingen dataset. (b) Unimodal Vaihingen dataset. (c) Multimodal DDHR dataset. (d) Unimodal DDHR dataset.

TABLE V

COMPARISON OF DIFFERENT METHODS IN TERMS OF COMPLEXITY, PARAMETERS, AND INFERENCE TIME. MIOU REPRESENTS THE MODEL PERFORMANCE ON THE ISPRS VAIHINGEN DATASET. BOLD VALUES INDICATE THE BEST RESULTS, AND UNDERLINED VALUES INDICATE THE SECOND-BEST RESULTS

Type	Method	FLOPs (G)	Param (M)	Inference Time (ms)	mIoU (%)
Unimodal	A2FPN [28]	41.83	22.82	**7.71**	79.26±0.44
	CMTFNet [35]	40.20	32.24	21.03	79.59±0.19
	RS3Mamba [38]	63.30	49.65	23.49	81.35±0.58
	RSAM-Seg [45]	85.87	112.09	80.69	80.31±0.18
	D2LS [56]	93.25	94.03	45.65	84.19±0.40
Multimodal	SAGate [51]	109.82	68.78	41.58	78.16±0.59
	CMX [57]	57.03	66.56	50.16	86.77±0.22
	PACSCNet [14]	150.21	89.96	48.68	87.41±0.10
	AMIANet [21]	73.74	59.38	41.72	87.81±0.20
	FTransUNet [19]	225.45	203.99	126.91	86.06±0.34
	MFNet [58]	113.13	96.93	266.57	85.27±0.53
	StitchFusion [20]	**15.10**	**18.17**	21.08	85.84±0.36
	CSFAFormer [52]	141.76	77.85	114.31	88.05±0.16
	R2SNet (Ours)	127.39	98.97	121.83	**89.06±0.33**

E. Computational Complexity and Efficiency Analysis

Computational complexity and efficiency are evaluated using FLOPs, Params, and inference time. As shown in Table V, R2SNet requires 127.39G FLOPs, 98.97M parameters, and 121.83 ms inference time. Although it is not the most lightweight model, its computational cost is still comparable among advanced multimodal methods. For example, R2SNet has lower FLOPs than PACSCNet, FTransUNet, and CSFAFormer, and lower inference time than FTransUNet and MFNet. In terms of performance, R2SNet achieves the best mIoU of 89.06% ± 0.33%. Compared with lightweight methods such as StitchFusion, R2SNet requires more computation but obtains higher segmentation accuracy. Compared with CSFAFormer, R2SNet achieves better mIoU with lower FLOPs. These results indicate that R2SNet provides a reasonable trade-off between segmentation accuracy and computational cost.

F. Ablation Study

We conducted a series of ablation studies to thoroughly analyze the implementation of R2SNet and evaluate the performance of core components. These studies were validated on the Vaihingen and DDHR datasets. In this discussion, we primarily focus on two performance metrics: m$F1$ and mIoU.

TABLE VI

MODAL ABLATION RESULTS ON THE VAIHINGEN AND DDHR DATASETS

Type	Vaihingen		DDHR	
	mF1 (%)	mIoU (%)	mF1 (%)	mIoU (%)
Unimodal	93.12±0.23	87.56±0.36	92.94±0.41	87.04±0.58
Multimodal	94.16±0.18	89.06±0.33	94.80±0.36	90.11±0.53

1) Contribution of Modality: To evaluate the contribution of different modalities and further analyze the information provided by auxiliary modalities in optical-DSM and optical-SAR settings, we conducted modality ablation experiments. Fig. 11 presents the confusion matrices from modality ablation experiments performed on the Vaihingen and DDHR datasets. Compared with unimodal optical RSI, the multimodal setting generally produces clearer diagonal responses in the confusion matrices, indicating improved category discrimination and reduced interclass confusion. This validates the positive role of auxiliary modalities in enhancing recognition performance.

Table VI reports the quantitative results of the modal ablation experiments. On the Vaihingen dataset, the multimodal setting improves m$F1$ and mIoU from 93.12% ±0.23% and 87.56% ±0.36% to 94.16% ±0.18% and 89.06% ±0.33%, yielding gains of 1.04% and 1.50%, respectively. On the DDHR dataset, m$F1$ and mIoU increase from 92.94% ±0.41% and 87.04% ±0.58% to 94.80% ±0.36% and 90.11% ±0.53%, corresponding to improvements of 1.86% and 3.07%. These results show that auxiliary modalities provide effective complementary information for both optical-DSM and optical-SAR segmentation.

The improvement is more evident on the DDHR dataset, where SAR provides scattering responses and structural texture cues that compensate for the limitations of optical imagery in low-resolution or texture-insufficient regions. In contrast, DSM mainly contributes elevation and geometric boundary information, which helps distinguish spectrally similar categories such as buildings, trees, and low vegetation. Overall, the results indicate that DSM and SAR contribute complementary information in different ways, and their integration improves semantic discrimination and boundary localization.

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TABLE VII
ABLATION RESULTS FOR THE CORE COMPONENTS ON THE VAIHINGEN AND DDHR DATASETS

Method	Vaihingen		DDHR	
	mF1 (%)	mIoU (%)	mF1 (%)	mIoU (%)
R2SNet	94.16±0.18	89.06±0.33	94.80±0.36	90.11±0.53
R2SNet w/o ADB	92.95±0.32	87.47±0.52	93.78±0.49	88.91±0.64
R2SNet w/o SGCF	93.12±0.19	87.63±0.35	93.88±0.36	89.02±0.56
R2SNet w/o PSR	92.99±0.25	87.50±0.41	93.82±0.43	88.96±0.58
Baseline	91.43±0.20	84.45±0.38	91.58±0.45	85.80±0.61
Baseline w/ ADB	92.07±0.18	86.21±0.29	92.63±0.34	86.97±0.52
Baseline w/ SGCF	91.73±0.34	85.52±0.58	92.56±0.47	86.51±0.64
Baseline w/ PSR	91.76±0.28	85.65±0.46	92.61±0.38	86.92±0.58

2) Effectiveness of Core Components: The quantitative results of the component ablation studies are presented in Table VII. Removed components are marked as (w/o), while added components are denoted as (w/). The baseline employs two independent multilayer convolutions for feature extraction, followed by feature fusion via standard cross-attention, and then connects to a segmentation head to produce segmentation results. When adding components, the corresponding module is directly inserted at the appropriate position before the segmentation head.

Starting from the baseline, adding each core module brings consistent performance gains. On the Vaihingen dataset, adding ADB improves mIoU from 84.45% ± 0.38% to 86.21% ± 0.29%, adding SGCF improves it to 85.52% ± 0.58%, and adding PSR improves it to 85.65% ± 0.46%. On the DDHR dataset, the corresponding mIoU values increase from 85.80% ± 0.61% to 86.97% ± 0.52%, 86.51% ± 0.64%, and 86.92% ± 0.58%, respectively. This confirms that each module contributes positively to the overall framework. When removing modules from the complete R2SNet, the performance consistently drops. Removing ADB reduces the mIoU from 89.06% ± 0.33% to 87.47% ± 0.52% on Vaihingen and from 90.11% ± 0.53% to 88.91% ± 0.64% on DDHR. Removing SGCF decreases the mIoU to 87.63% ± 0.35% and 89.02% ± 0.56%, while removing PSR decreases it to 87.50% ± 0.41% and 88.96% ± 0.58%, respectively. These results indicate that ADB, SGCF, and PSR play complementary roles in feature extraction, cross-modal fusion, and structural refinement.

Fig. 12 validates the contribution of the ADB encoder to the robust initial feature representation capability of R2SNet. After removing the ADB encoder, fragmented segmentation and incomplete boundaries emerge in capturing the complete structure and details of the building. R2SNet effectively mitigates redundancy and modal interference by performing modality-specific extraction of semantic and geometric features through the ADB encoder, enabling accurate and coherent segmentation. Fig. 13 reveals the critical role of the SGCF module in guiding cross-modal feature fusion. Without the SGCF module, segmentation results exhibit inconsistencies and misclassifications. This indicates that effectively integrating heterogeneous information and suppressing unreasonable cross-modal associations remains a key challenge. The complete R2SNet successfully embeds spatial semantic priors into the fusion process, achieving clear and accurate results. Fig. 14 elucidates the necessity of the PSR decoder for refining details and generating structurally complete segmentations. Removing the PSR decoder degrades boundary precision and structural integrity, resulting in rough and inaccurate land cover boundaries. The complete R2SNet achieves clear, smooth, and precise boundaries by leveraging multiscale aggregation and eight-directional selective scanning.

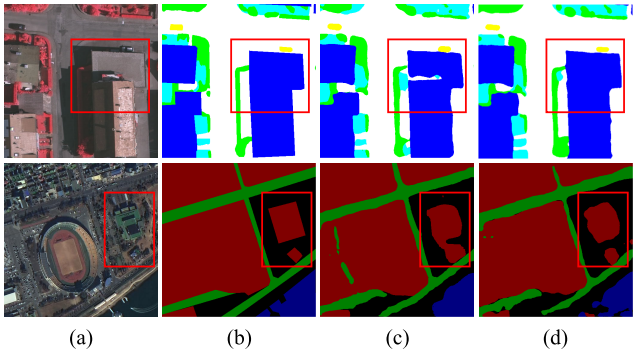


Fig. 12. Visualization results of ablation studies. (a) Input optical RSI. (b) Ground truth. (c) R2SNet without ADB encoder. (d) R2SNet.

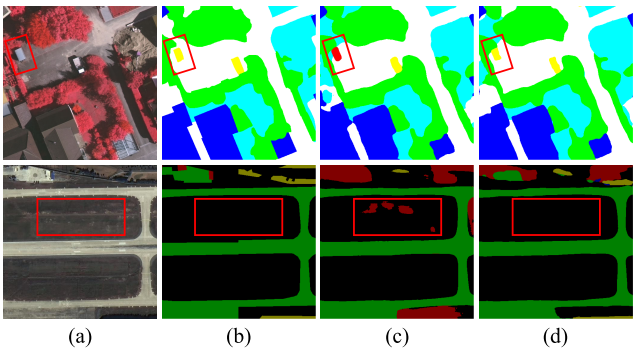


Fig. 13. Visualization results of ablation studies. (a) Input optical RSI. (b) Ground truth. (c) R2SNet without SGCF module. (d) R2SNet.

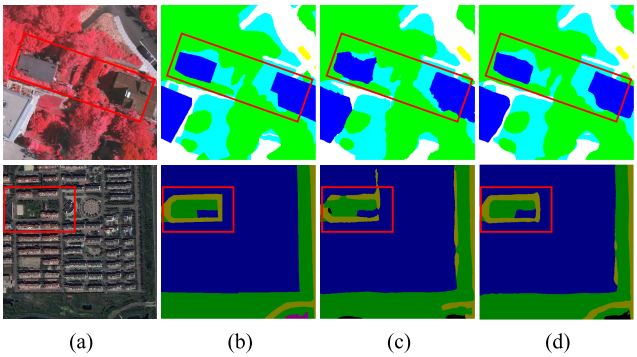


Fig. 14. Visualization results of ablation studies. (a) Input optical RSI. (b) Ground truth. (c) R2SNet without PSR decoder. (d) R2SNet.

3) Feature Flow Visualization of R2SNet: Fig. 15 visualizes the feature flow across different stages of R2SNet, illustrating the progressive process of multimodal feature extraction, fusion, and refinement. Fig. 15(a) displays the input optical RSI, while Fig. 15(b) and (c) show the features processed by the geometric branch and semantic branch of the ADB

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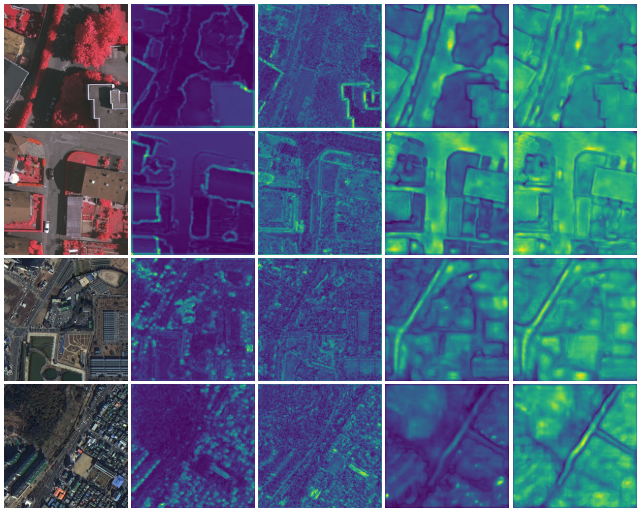


Fig. 15. Feature visualizations for each stage of R2SNet are presented. (a) Input optical RSI. (b) Features after the geometric branch of the ADB encoder. (c) Features after the semantic branch of the ADB encoder. (d) Features after the SGCF module. (e) Features after the PSR decoder.

TABLE VIII
ABLATION RESULTS OF THE ADB ENCODER ON THE VAIHINGEN AND DDHR DATASETS

Encode Branch	Vaihingen		DDHR		FLOPs
	mF1	mIoU	mF1	mIoU	
w/o Adapter	93.44±0.40	88.07±0.60	93.62±0.24	88.51±0.41	125.79
Semantic Only	92.15±0.17	84.82±1.23	92.28±0.32	85.50±1.78	209.61
Geometric Only	93.24±0.19	87.46±0.34	93.20±0.21	88.25±0.32	59.41
Branch Swap	90.15±0.77	82.76±1.27	90.49±0.86	83.16±1.67	141.63
ADB	94.16±0.18	89.06±0.33	94.80±0.36	90.11±0.53	127.39

encoder, respectively. After the SGCF module, as shown in Fig. 15(d), the fused features become more semantically organized, suggesting that spatial-semantic guidance helps align cross-modal information. This observation suggests that spatial-semantic guidance can help regularize cross-modal fusion under moderate modality inconsistencies, while severe modality degradation may still affect the reliability of the guidance mechanism. Fig. 15(e) further shows that the features after the PSR decoder exhibit clearer structural responses and stronger foreground-background contrast. This feature evolution matches the design motivation of R2SNet, progressing from modality-specific representation to spatial-semantic fusion and structure refinement.

4) Analysis of the ADB Encoder: The ADB encoder assigns the semantic branch to the optical modality and the geometric branch to the auxiliary modality. To verify this design, Table VIII compares four variants: removing the Adapter, using the semantic branch for both modalities, using the geometric branch for both modalities, and swapping the two branches.

Removing the Adapter decreases mIoU from 89.06% ± 0.33% to 88.07% ± 0.60% on Vaihingen and from 90.11% ± 0.53% to 88.51% ± 0.41% on DDHR, indicating its role in adapting pretrained semantic features to remote sensing scenes. Using the semantic branch for both modalities brings the highest FLOPs of 209.61G but yields limited mIoU, while using the geometric branch for both modalities reduces FLOPs to 59.41G but also weakens performance. The Swap setting obtains the lowest mIoU, 82.76% ± 1.27% on Vaihingen and 83.16% ± 1.67% on DDHR, showing that improper branch assignment harms feature extraction. In contrast, the complete ADB achieves the best mIoU of 89.06% ± 0.33% and 90.11% ± 0.53%, demonstrating that the asymmetric semantic-geometric design better exploits modality characteristics for subsequent fusion and refinement.

TABLE IX
ABLATION RESULTS OF DIFFERENT MULTIMODAL FUSION STRATEGIES ON THE VAIHINGEN AND DDHR DATASETS

Fusion Strategy	Vaihingen		DDHR	
	mF1 (%)	mIoU (%)	mF1 (%)	mIoU (%)
Add	92.94±0.31	86.96±0.52	93.47±0.49	88.32±0.60
Concat	93.09±0.22	87.28±0.39	93.58±0.39	88.62±0.52
CA	93.12±0.19	87.63±0.35	93.88±0.36	89.02±0.56
SGCF	94.16±0.18	89.06±0.33	94.80±0.36	90.11±0.53

5) Analysis of the SGCF Module: The SGCF module is designed to incorporate spatial-semantic priors into cross-modal feature fusion. By encoding semantic distributions and spatial positions into a guidance bias, SGCF constrains cross-modal attention and reduces unreasonable semantic associations.

As shown in Table IX, SGCF achieves the best results among the compared fusion strategies. On the Vaihingen dataset, SGCF obtains 94.16% ± 0.18% m$F1$ and 89.06% ± 0.33% mIoU, improving over standard cross-attention by 1.04% and 1.43%, respectively. Compared with Add and Concat fusion, SGCF also shows consistent improvements on both datasets, indicating that explicitly modeling spatial-semantic relationships is beneficial for multimodal fusion. The qualitative results in Fig. 13 show that removing SGCF causes more inconsistent predictions and local misclassifications.

Fig. 16 further visualizes the conditional category relationship matrices. To compute each matrix, we first convert the ground truth or predicted segmentation map into class occurrence indicators at the image patch level. For a category c_i, the matrix element R_{ij} is calculated as the conditional occurrence probability of category c_j when c_i appears, namely $R_{ij} = P(c_j|c_i)$. Thus, larger values indicate stronger category co-occurrence, while smaller values indicate weaker association or potential mutual exclusion. Compared with the model without SGCF, R2SNet produces matrices closer to the ground truth, especially for category pairs with clear spatial dependence or weak association. This suggests that SGCF helps preserve more reasonable category relationship patterns in the predicted segmentation maps.

6) Analysis of the PSR Decoder: After multimodal features are extracted by ADB and fused by SGCF, the PSR decoder further aggregates multiscale features and refines spatial structures. It promotes cross-scale interaction through progressive

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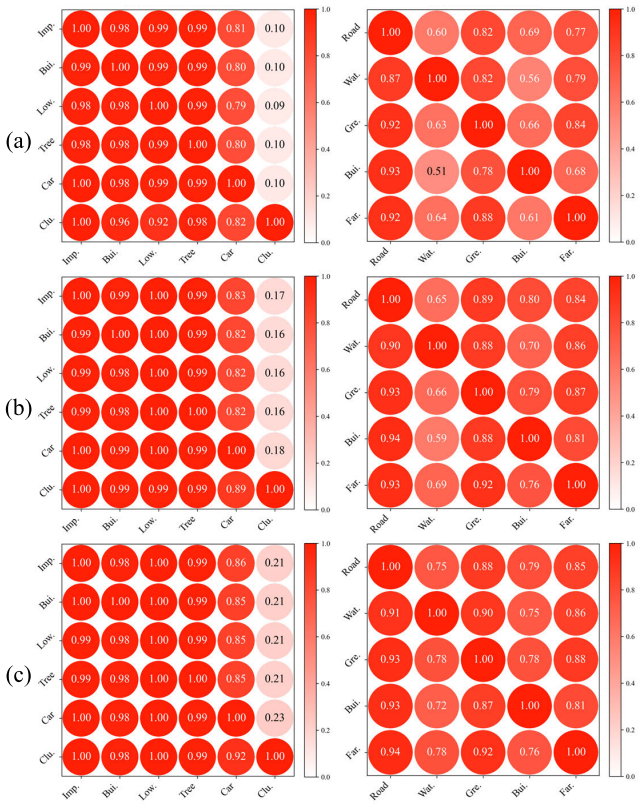


Fig. 16. Visualization of conditional category relationship matrices. The first and second rows show the results on the Vaihingen and DDHR datasets, respectively. A predicted matrix that is more consistent with the ground truth indicates better preservation of true category co-occurrence and mutual-exclusion relationships. (a) Ground truth. (b) R2SNet. (c) R2SNet w/o SGCF.

TABLE X
ABLATION RESULTS OF THE PSR DECODER WITH DIFFERENT SCANS ON THE VAIHINGEN AND DDHR DATASETS

Type	Vaihingen		DDHR	
	mF1 (%)	mIoU (%)	mF1 (%)	mIoU (%)
4-SS2D	93.24±0.20	87.68±0.34	94.19±0.42	89.38±0.57
8-SS2D	94.16±0.18	89.06±0.33	94.80±0.36	90.11±0.53

multiscale fusion and enhances directional structural modeling through selective scanning. Table X compares four-directional SS2D and the proposed eight-directional scanning strategy. Compared with 4-SS2D, 8-SS2D improves m$F1$ and mIoU by 0.92% and 1.38% on the Vaihingen dataset, respectively. On the DDHR dataset, it brings moderate gains of 0.61% in m$F1$ and 0.73% in mIoU. These consistent improvements indicate that diagonal scanning directions provide useful complementary structural information. Fig. 14 shows that removing the PSR decoder leads to less complete boundaries and weaker structural continuity. With PSR, R2SNet better preserves fine boundaries and linear structures, especially for buildings, roads, and irregular land cover boundaries. This suggests that PSR further refines the features generated by SGCF in the spatial domain, allowing semantic constraints to be reflected in more continuous boundaries and more complete land cover structures.

V. LIMITATIONS

The proposed R2SNet indicates competitive and robust performance across several datasets, yet certain limitations remain to be further explored. The current model retains a relatively high computational cost. The frozen MobileSAM backbone helps control the encoder scale, while the spatial-semantic attention in SGCF and the multidirectional structure refinement in PSR still introduce additional memory consumption and inference latency. This may limit its deployment in real-time or resource-constrained remote sensing applications. In optical-SAR scenarios, SAR images may be affected by speckle noise, side-looking imaging geometry, geometric distortions, and local registration errors. Moreover, since R2SNet relies on spatial-semantic priors and multimodal inputs, inaccurate auxiliary modalities, inconsistent multimodal registration, or highly complex scene distributions may affect the reliability of the learned relationship guidance. Furthermore, variations in imaging conditions, modality quality, and sensor noise levels may change the relative reliability of optical and auxiliary modalities, which can influence the robustness of the spatial-semantic guidance mechanism during cross-modal fusion. Therefore, explicitly modeling SAR geometric distortions and estimating modality reliability remain important directions for future research. In addition, R2SNet has only been validated on semantic segmentation, while its potential for other dense prediction tasks, such as object detection, instance segmentation, and change detection, remains to be further explored.

VI. CONCLUSION

This article proposes a novel multimodal RSI semantic segmentation network, R2SNet, which establishes a progressive strategy from semantic relationship modeling to spatial structure refinement. This method addresses the challenges of overlooked coexistence and mutual-exclusion relationships, as well as fine-grained boundary segmentation. The ADB encoder enables modality-specific feature extraction, mitigating modal interference and computational redundancy. The SGCF module explicitly embeds spatial-semantic priors, integrating land cover relationships into cross-modal feature fusion to suppress unreasonable associations and enhance semantic consistency. The PSR decoder aggregates multiscale features and further refines them to improve boundary precision and structural integrity. Extensive experiments on four benchmark datasets show that R2SNet achieves competitive performance in semantic consistency, boundary refinement, and robustness to complex scenes. Ablation studies and visual analysis further verify the complementary contributions of modality-specific feature extraction, spatial-semantic guidance, and structural refinement. Overall, R2SNet provides an effective framework for multimodal RSI segmentation and offers useful insight into land cover relationship modeling. Future work will explore more efficient feature modeling, modality reliability estimation, and distortion-aware alignment to improve the efficiency and adaptability of R2SNet under complex multimodal imaging conditions, and further extend it to broader dense prediction tasks.

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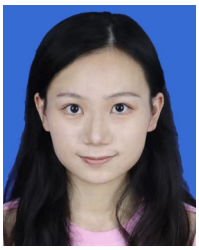
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